# SYSTEM DESIGN

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# Edu-Connect

Learning Management System


Prepared by:

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# SYSTEM DESIGN

## Introduction

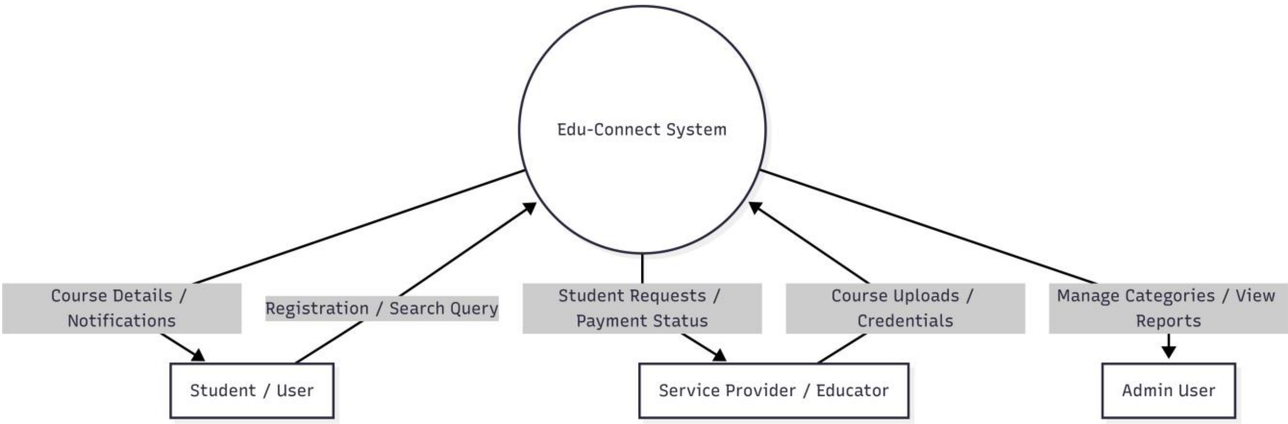
Software design is a process through which the requirements gathered during analysis are translated into a structured representation of the software system. In detailed design, we specify how the individual modules within the system interact with each other and define the internal processing logic of each component. Detailed specifications are provided in a manner that clearly describes what each module is supposed to do, how it handles inputs, and what outputs it produces.

Detail design specification describes the complete features of the Edu-Connect system. It is the refinement of the system design that essentially expands high-level architecture to contain more detailed descriptions of the processing logic of components and data structures such that the design is sufficiently complete to be implemented by developers. This phase includes context diagrams, data flow diagrams, entity-relationship diagrams, normalized database tables, and modular decomposition of each role in the system.

## Context Flow Diagram

A Context Flow Diagram is a top-level data flow diagram. It contains only one process node that generalizes the function of the entire system in relation to external entities. In the context diagram, the entire Edu-Connect system is treated as a single process and all its inputs, outputs, sinks, and sources are identified and shown. The external entities for this system are the Student/User, the Service Provider (Teacher/Educator), and the Admin User.

The diagram below represents the Context Flow Diagram of the Edu-Connect System:




## Data Flow

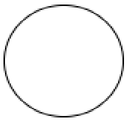
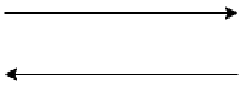
### Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation of the system or a portion of the system. It consists of data flows, processes, sources, sinks, and data stores, all described through the use of well-defined symbols. It illustrates how data enters the system, moves between processes, and exits. DFDs are used to communicate system design between technical and non-technical stakeholders and serve as the foundation for detailed module specifications.

**Rules Regarding DFD Construction:**

- A process cannot have only outputs.
- A process cannot have only inputs.
- The inputs to a process must be sufficient to produce the outputs from the process.
- All data stores must be connected to at least one process.
- All data stores must be connected to a source or sink.
- A data flow can have only one direction of flow. Multiple data flows to and/or from the same process and data store must be shown by separate arrows.
- If the exact same data flows to two separate destinations, it should be represented by a forked arrow.
- Data cannot flow directly back into the process it has just left.
- All data flows must be named using a noun phrase.

**DFD Symbols:**

| Name | Notation | Description |
|---|---|---|
| Process |  | A process transforms incoming data flow into outgoing data flow. Processes are shown by named circles (bubbles). |
| Datastore |  | Data stores are repositories of data in the system. They are sometimes also referred to as files or databases. |
| Dataflows |  | Data flows are pipelines through which packets of information flow. Arrows are labelled with the name of the data that moves through them. |
| External Entity |  | External entities are objects outside the system with which the system communicates. They are sources and destinations of the system's inputs and outputs. |

## DFD Level 1 for Admin:

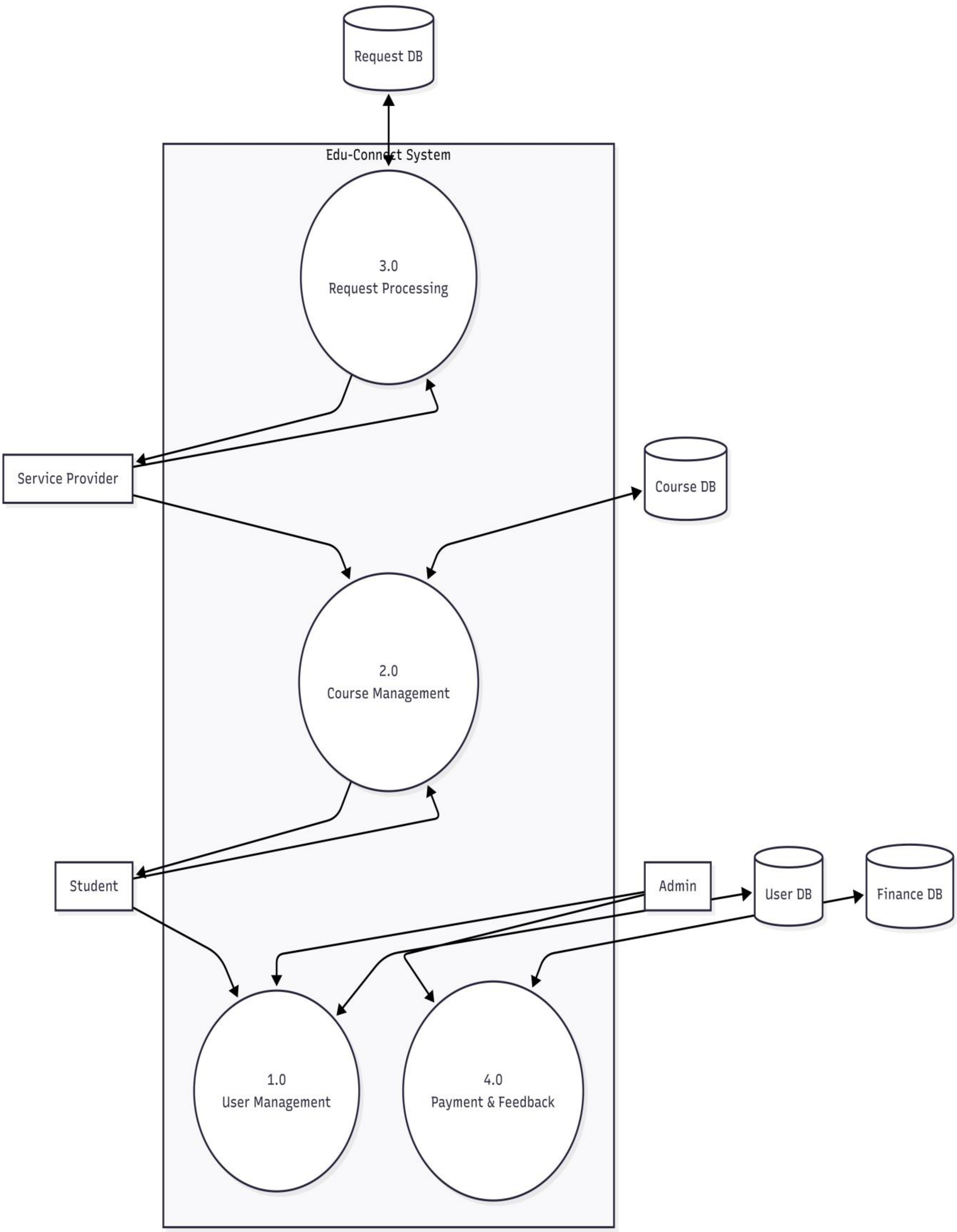
The Level 1 DFD for the Admin role decomposes the system functions handled by the administrator. The Admin interacts with the Edu-Connect system through the following processes: Login (with username and password validated against the admin datastore), User Management (viewing and approving or rejecting student and teacher registrations), Material Management (monitoring and deleting uploaded learning materials), Notification Management (composing and dispatching notifications to target user groups), and Approval Processing (recording approval decisions along with remarks and timestamps). Data flows from the Admin entity into each process and the corresponding datastores — the Users DB, Materials DB, Notifications DB, and Approvals DB — return confirmation data or records back to the Admin.

## DFD level 1 for Service Provider (Teacher):

The Level 1 DFD for the Teacher (Service Provider) role decomposes the processes specific to an educator on the platform. The Teacher interacts with the system through: Registration (submitting credentials, college, and subject details to the Admin datastore pending approval), Login (username and password verification against the Users DB), Upload Materials (submitting PDFs, videos, images, or external links to the Materials DB), Send Notification (dispatching study-related announcements to the Notifications DB targeting students), View Profile (reading and updating personal information), and View/Submit Feedback (submitting ratings and comments to the Feedback DB). Relevant data flows — such as material IDs, notification titles, and approval statuses — travel between the Teacher entity and the system datastores.

## DFD level 1 for User (Student):

The Level 1 DFD for the Student (User) role captures the processes available to a registered and approved learner. The Student entity interacts with the system through: Registration (providing personal details and class information which flow into the Users DB pending admin approval), Login (submitting email and password for verification), Browse & Search Materials (querying the Materials DB by keyword, file type, or uploader to retrieve relevant study resources), View Notifications (reading notifications from the Notifications DB and marking them as read via the Notification Reads datastore), View & Update Profile (reading and editing personal information), and Submit Feedback (rating and commenting on the platform or materials via the Feedback DB). Each process consumes specific input data flows and produces corresponding output data flows back to the Student entity.

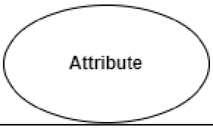
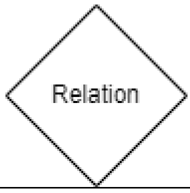

Request DB
Edu-Connect System
3.0
Request Processing
Service Provider
Course DB
2.0
Course Management
Student
Admin
User DB
Finance DB
1.0
User Management
4.0
Payment & Feedback

# Entity-Relationship Diagram

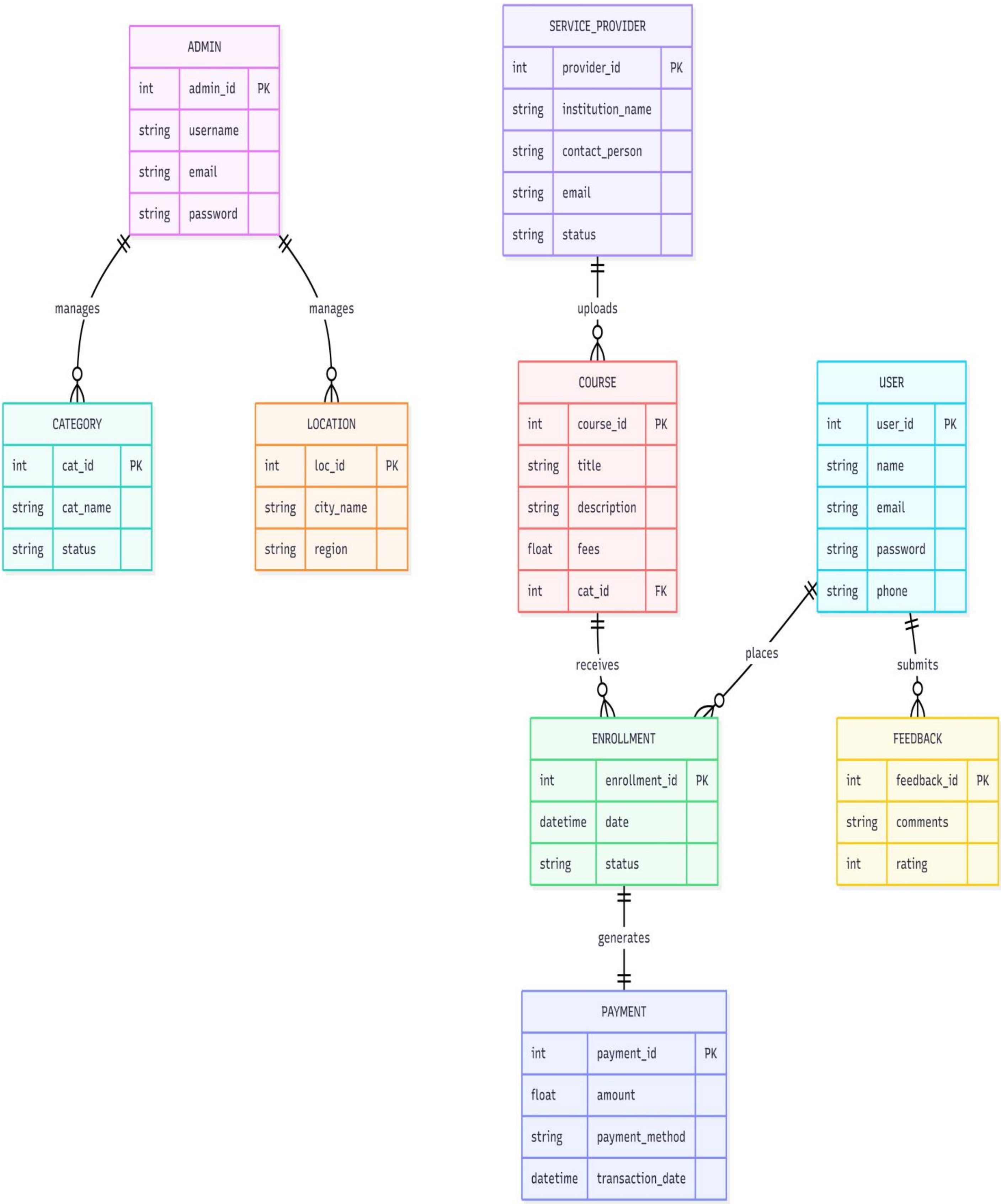
## Introduction

The basic objective of the ER model representation is to identify entities which are 'things' in the real world with an independent existence. Entities are physical items or aggregations of data items that are important to the business we analyze or to the system we intend to build. An entity represents an object defined within the information system about which we want to store information. Entities are named as singular nouns and are shown in rectangles in an ER Diagram. Each entity is described by several attributes; individual instances of an entity will have different attribute values. For the Edu-Connect system, the primary entities are User, Student, Teacher, Admin, Materials, Notifications, Notification Reads, Approvals, Study Videos, Video Likes, Video Comments, Video Saves, and Feedback.

### ER Diagram Symbols:

| Name | Notation | Description |
| --- | --- | --- |
| Entity |   | It may be an object with physical or conceptual existence. Represented by a Rectangle. |
| Attribute |   | The properties of the entity. Represented by an Ellipse. |
| Relationship |   | Whenever an attribute of one entity refers to another entity, a relationship exists. Represented by a Diamond. |
| Link | | Lines link attributes to entity sets and entity sets to relationships. |
| Derived Attribute |   | A dashed ellipse denotes a derived attribute whose value can be computed from other attributes. |
| Key Attribute |   | An entity type usually has an attribute whose values are distinct for each individual entry in the entity set. Represented by an underlined word in an ellipse. |
| Cardinality Ratio | 1:1, 1:M, M:1, M:M | It specifies the maximum number of relationship instances that an entity can participate in. There are four cardinality ratios: one-to-one, one-to-many, many-to-one, and many-to-many. |

### ER Diagram:


ADMIN
int
admin_id
PK
string
username
string
email
string
password
manages
manages
CATEGORY
int
cat_id
PK
string
cat_name
string
status
LOCATION
int
loc_id
PK
string
city_name
string
region
SERVICE_PROVIDER
int
provider_id
PK
string
institution_name
string
contact_person
string
email
string
status
uploads
COURSE
int
course_id
PK
string
title
string
description
float
fees
int
cat_id
FK
USER
int
user_id
PK
string
name
string
email
string
password
string
phone
receives
places
submits
ENROLLMENT
int
enrollment_id
PK
datetime
date
string
status
FEEDBACK
int
feedback_id
PK
string
comments
int
rating
generates
PAYMENT
int
payment_id
PK
float
amount
string
payment_method
datetime
transaction_date

# DATABASE DESIGN

## Introduction

The term 'database' is used to describe everything from a single set of data to a complex set of tools such as MySQL, and a whole lot in between. The term data model is used to mean the conceptual description of the problem space. This includes the definition of entities, their attributes, and the entity constraints. The data model also includes a description of the relationships between entities and any constraints on those relationships. It is the translation of the conceptual model into a physical representation which shall be implemented using a database management system. The Edu-Connect system uses a MySQL relational database hosted via XAMPP. The main advantage of this approach is to reduce manual data handling. The system performs required data operations, maintains consistent and error-free records, and validates all transactions within a short period of time.

The Database Management System (DBMS) provides flexibility in data storage and retrieval and in the production of information. It determines what type of data is needed and how it is processed, while the operating system is responsible for placing the data on storage devices. A schema defines the database structure, and the sub-schema defines the portion of the database that a specific program or module will use.

### Data Retrieval Process:

- The application module determines what data is needed and communicates the request to the DBMS.
- The DBMS verifies that the requested data is in fact stored in the database.
- The data must be defined within the appropriate sub-schema for the requesting module.
- The DBMS instructs the operating system to locate and retrieve the data from the specific location on the disk.
- A copy of the data is passed to the application layer for processing and presentation.
- A database must be created and populated before it can be used for queries or operations.

## Normalized Tables:

### Table Name: users

Description: Base authentication and profile table storing details of all system users — students, teachers, and admin.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| user_id | INT | Primary Key, Auto Increment | Unique identifier for each user |
| unique_id | VARCHAR(20) | NOT NULL, UNIQUE | System-generated role-based ID (e.g., STU-000001) |
| name | VARCHAR(100) | NOT NULL | Full name of the user |
| email | VARCHAR(150) | NOT NULL, UNIQUE | Email address used for login |
| password | VARCHAR(255) | NOT NULL | Hashed password for authentication |
| role | ENUM | NOT NULL, DEFAULT 'student' | Role of the user: student, teacher, or admin |
| college_name | VARCHAR(200) | NULL | Name of the institution the user belongs to |
| contact_number | VARCHAR(15) | NULL | Phone number of the user |
| age | INT | NULL | Age of the user |
| is_approved | ENUM | DEFAULT 'pending' | Account approval status: pending, approved, or rejected |
| profile_photo | VARCHAR(255) | NULL | File path to the user's profile photograph |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Record creation timestamp |
| updated_at | TIMESTAMP | ON UPDATE CURRENT_TIMESTAMP | Record last updated timestamp |

### Table Name: students

Description: Stores additional profile information specific to student users.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| student_id | INT | Primary Key, Auto Increment | Unique identifier for student record |
| user_id | INT | Foreign Key → users(user_id) | References the base user record |
| class_name | VARCHAR(100) | NULL | Class or semester the student is enrolled in |

### Table Name: teachers

Description: Stores additional profile information specific to teacher users.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| teacher_id | INT | Primary Key, Auto Increment | Unique identifier for teacher record |
| user_id | INT | Foreign Key → users(user_id) | References the base user record |
| subject | VARCHAR(100) | NULL | Subject or specialization the teacher handles |

### Table Name: admin

Description: Stores the admin record linked to the base users table.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| admin_id | INT | Primary Key, Auto Increment | Unique identifier for admin record |
| user_id | INT | Foreign Key → users(user_id) | References the base user record |

### Table Name: materials

Description: Stores details of learning materials uploaded by teachers, including PDFs, videos, images, and links.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| material_id | INT | Primary Key, Auto Increment | Unique identifier for each material |
| user_id | INT | Foreign Key → users(user_id) | ID of the teacher who uploaded the material |
| title | VARCHAR(200) | NOT NULL | Title of the material |
| description | TEXT | NULL | Brief description of the material content |
| file_type | ENUM | NOT NULL | Type of material: pdf, video, image, or link |
| file_path | VARCHAR(500) | NULL | Server path to the uploaded file |
| external_link | VARCHAR(500) | NULL | External URL if the material is a link resource |
| upload_date | TIMESTAMP | DEFAULT CURRENT _TIMESTAMP | Date and time of upload |

### Table Name: notifications

Description: Stores notifications sent by admin or teachers to specific user groups.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| notification_id | INT | Primary Key, Auto Increment | Unique identifier for each notification |
| sent_by | INT | Foreign Key → users(user_id) | ID of the user (admin or teacher) who sent the notification |
| title | VARCHAR(200) | NOT NULL | Title of the notification |
| message | TEXT | NOT NULL | Body content of the notification message |
| target_role | ENUM | DEFAULT 'all' | Target audience: all, student, or teacher |
| sender_role | ENUM | DEFAULT 'admin' | Role of the sender: admin or teacher |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Date and time when the notification was sent |

### Table Name: notification_reads

Description: Tracks which users have read which notifications to manage unread counts.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| id | INT | Primary Key, Auto Increment | Unique identifier for each read record |
| notification_id | INT | Foreign Key → notifications(notification_id) | Reference to the notification that was read |
| user_id | INT | Foreign Key → users(user_id) | Reference to the user who read the notification |
| read_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Timestamp when the notification was marked as read |

### Table Name: approvals

Description: Logs admin decisions on user registration requests including approval, rejection, and remarks.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| approval_id | INT | Primary Key, Auto Increment | Unique identifier for each approval record |
| user_id | INT | Foreign Key → users(user_id) | Reference to the user whose registration was reviewed |
| admin_id | INT | Foreign Key → users(user_id) | Reference to the admin who took the action |
| action | ENUM | NOT NULL | Action taken: approved or rejected |

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| remarks | TEXT | NULL | Optional remarks or reason provided by the admin |
| action_date | TIMESTAMP | DEFAULT CURRENT _TIMESTAMP | Date and time of the approval action |

**Table Name: study_videos**

Description: Stores YouTube video links shared by teachers as study resources.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| video_id | INT | Primary Key, Auto Increment | Unique identifier for each video entry |
| user_id | INT | Foreign Key → users(user_id) | ID of the teacher who shared the video |
| title | VARCHAR(200) | NOT NULL | Title of the video |
| description | TEXT | NULL | Description or summary of the video content |
| youtube_url | VARCHAR(500) | NOT NULL | Full YouTube URL of the video |
| youtube_id | VARCHAR(20) | NOT NULL | Extracted YouTube video ID for embedding |
| created_at | TIMESTAMP | DEFAULT CURRENT _TIMESTAMP | Date and time of submission |

### Table Name: video_likes

Description: Tracks likes given by users on study videos.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| like_id | INT | Primary Key, Auto Increment | Unique identifier for each like |
| video_id | INT | Foreign Key → study_videos(video_id) | Reference to the video that was liked |
| user_id | INT | Foreign Key → users(user_id) | Reference to the user who liked the video |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Timestamp of the like action |

### Table Name: video_comments

Description: Stores comments submitted by users on study videos.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| comment_id | INT | Primary Key, Auto Increment | Unique identifier for each comment |
| video_id | INT | Foreign Key → study_videos(video_id) | Reference to the commented video |
| user_id | INT | Foreign Key → users(user_id) | Reference to the user who commented |
| comment | TEXT | NOT NULL | Text content of the comment |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Timestamp of the comment submission |

### Table Name: video_saves

Description: Tracks videos bookmarked or saved by users for later reference.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| save_id | INT | Primary Key, Auto Increment | Unique identifier for each save record |
| video_id | INT | Foreign Key → study_videos(video_id) | Reference to the saved video |
| user_id | INT | Foreign Key → users(user_id) | Reference to the user who saved the video |
| created_at | TIMESTAMP | DEFAULT CURRENT_TIMESTAMP | Timestamp of the save action |

### Table Name: feedback

Description: Stores feedback, ratings, and comments submitted by users about the platform, materials, or teachers.

| Column Name | Data Type | Constraints | Description |
|---|---|---|---|
| feedback_id | INT | Primary Key, Auto Increment | Unique identifier for each feedback entry |
| user_id | INT | Foreign Key → users(user_id) | Reference to the user submitting feedback |
| target_type | ENUM | DEFAULT 'platform' | Target of feedback: platform, material, or teacher |
| target_id | INT | NULL | ID of the specific material or teacher being reviewed |
| rating | TINYINT | NOT NULL, CHECK 1-5 | Numerical rating between 1 and 5 |
| comment | TEXT | NULL | Written comment accompanying the rating |
| created_at | TIMESTAMP | DEFAULT CURRENT _TIMESTAMP | Date and time feedback was submitted |

## DETAILED DESIGN

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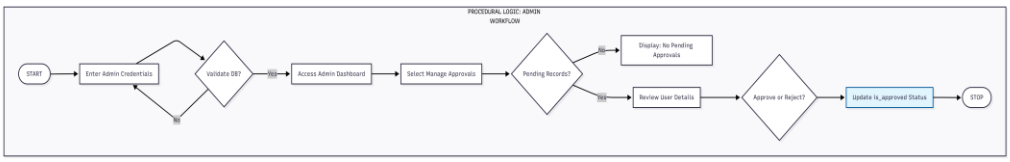
### Introduction

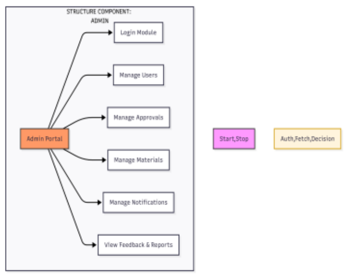
Detailed design is the second level of the design process. During detailed design, we specify how the modules in the Edu-Connect system interact with each other and the internal logic of each module specified during system design is decided. It is therefore also referred to as logic design. Detailed design essentially expands the system design and database design to contain a more detailed description of the processing logic and data structures so that the design is sufficiently complete for coding. Detailed design is the phase where the design is refined and plans, specifications, and module decompositions are created. This phase is where the complete functional scope of the project is identified and documented for implementation.

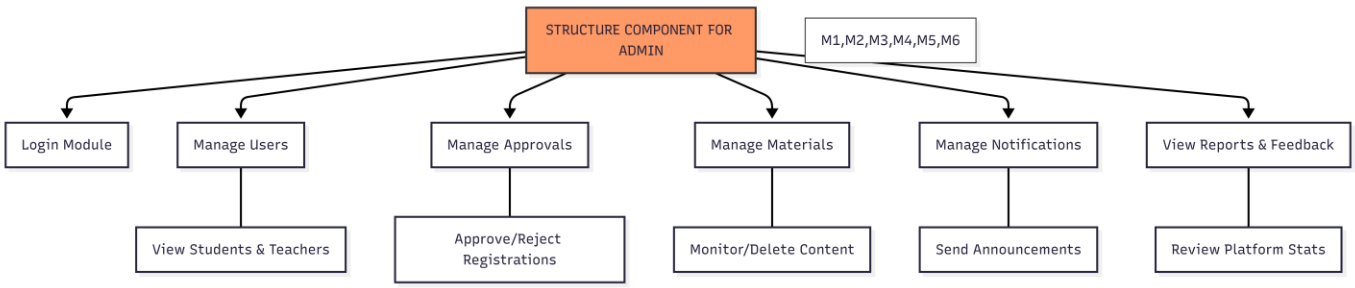
## I. STRUCTURE OF THE SOFTWARE PACKAGE

The functional components identified during the system design of Edu-Connect are organized into three major structural components based on the role of the user interacting with the system: Admin, Teacher (Service Provider), and Student (User). Each component contains sub-modules that handle specific responsibilities.

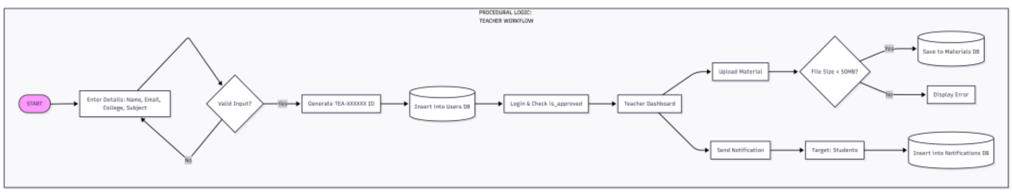
### STRUCTURE COMPONENT FOR ADMIN

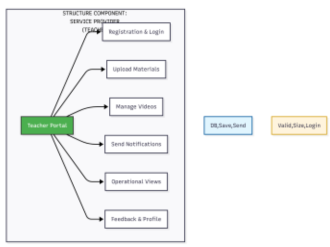


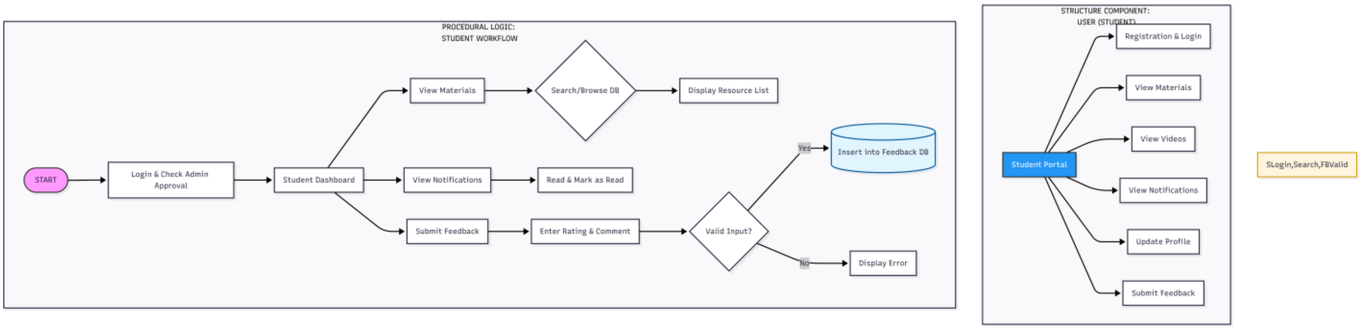




### STRUCTURE COMPONENT FOR SERVICE PROVIDER (TEACHER)



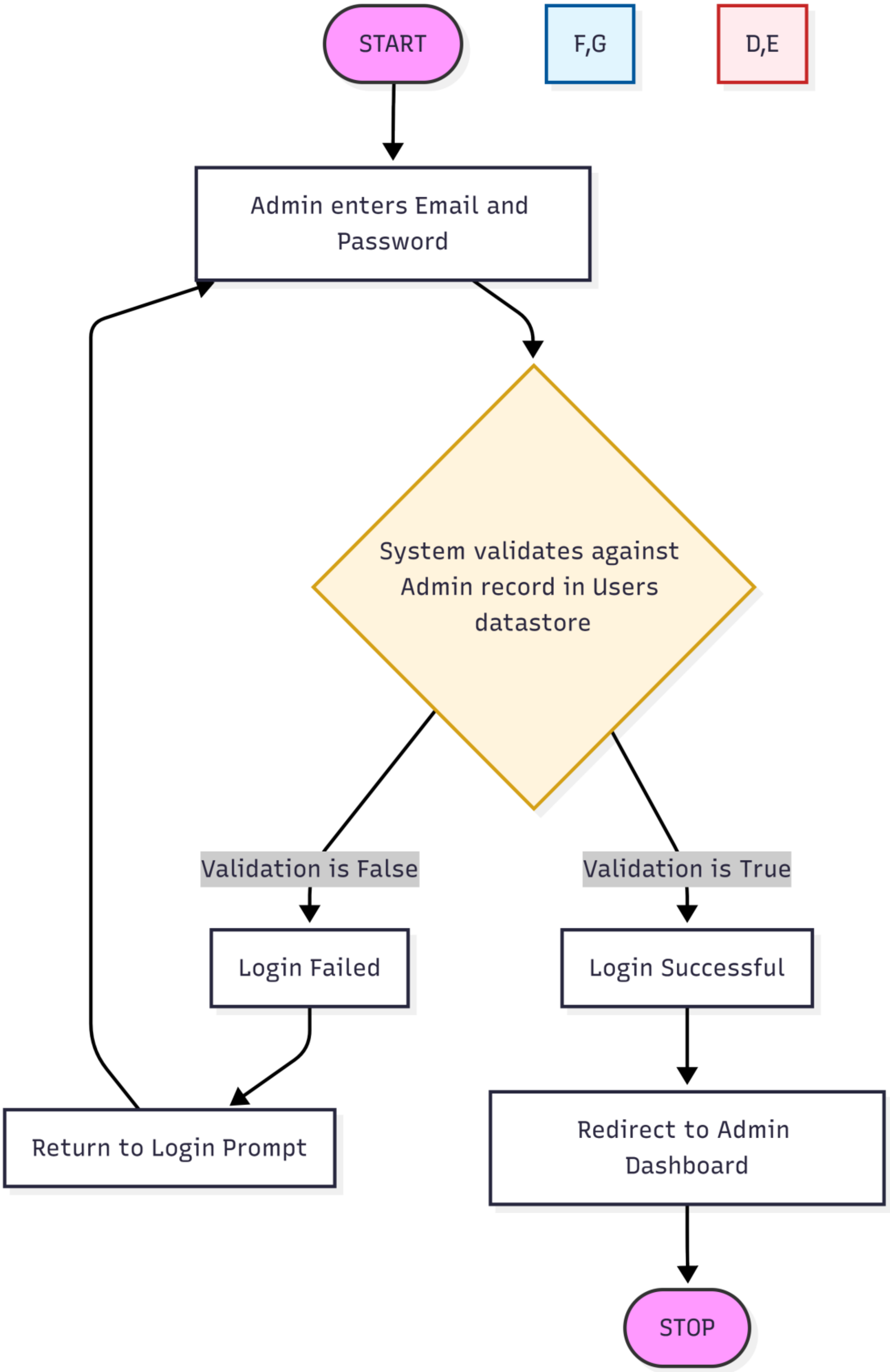

### STRUCTURE COMPONENT FOR USER (STUDENT)


PROCEDURAL LOGIC:
STUDENT WORKFLOW
START
Login & Check Admin Approval
Student Dashboard
View Materials
Search/Browse DB
Display Resource List
View Notifications
Read & Mark as Read
Submit Feedback
Enter Rating & Comment
Valid Input?
Yes
Insert into Feedback DB
No
Display Error
STRUCTURE COMPONENT:
USER (STUDENT)
Student Portal
Registration & Login
View Materials
View Videos
View Notifications
Update Profile
Submit Feedback
SLogin,Search,FBValid

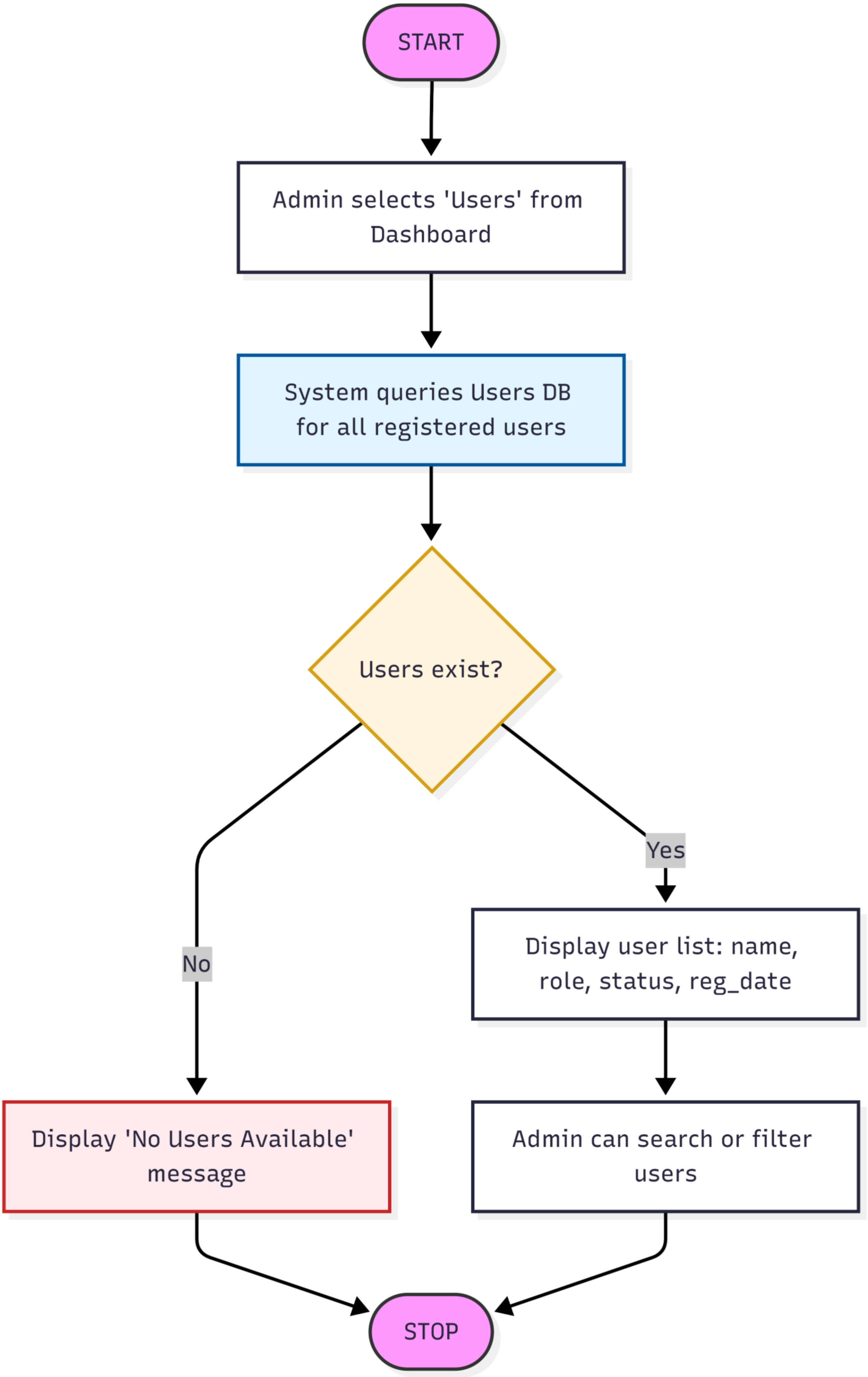
## MODULAR DECOMPOSITION COMPONENTS

### a. ADMIN:

**Login: Procedural Details**

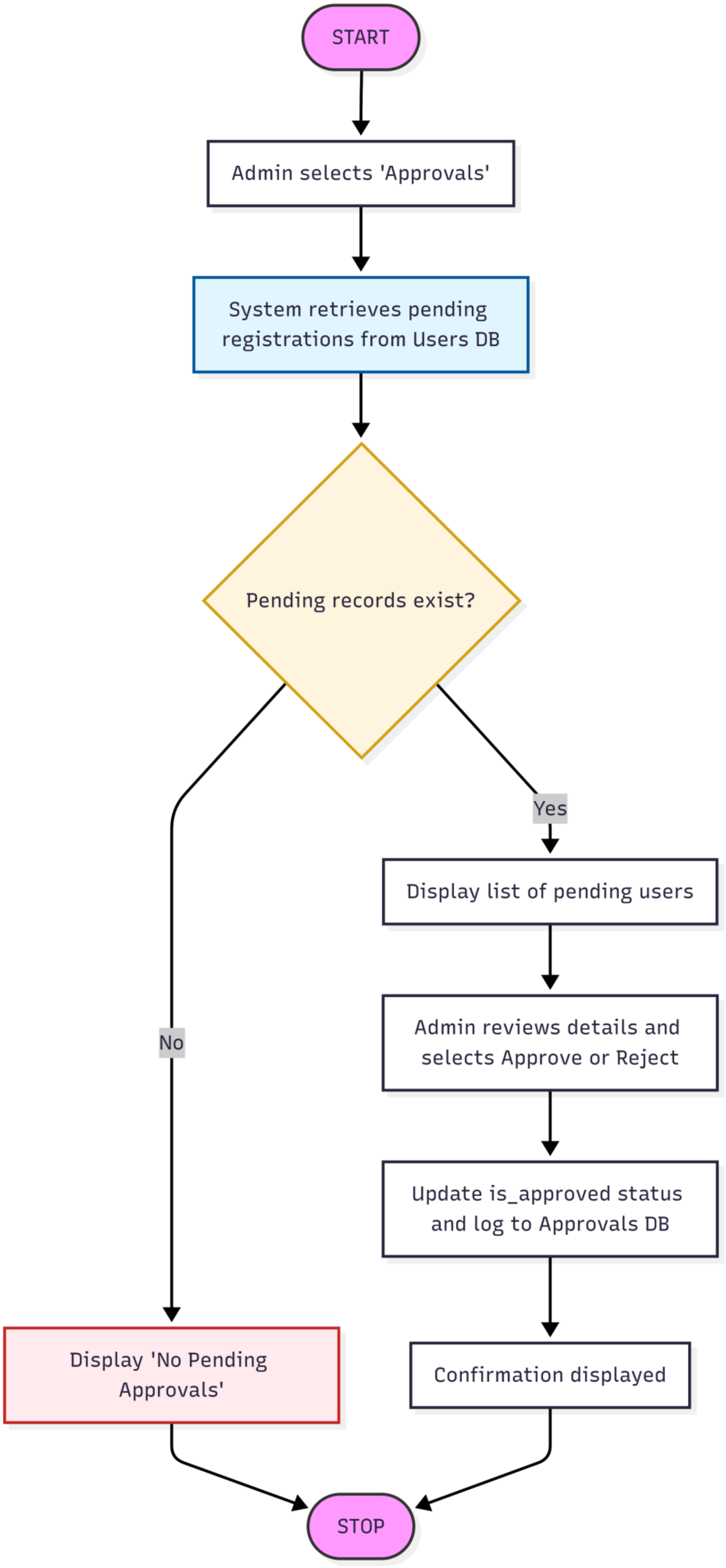



**Manage Users: Procedural Details**

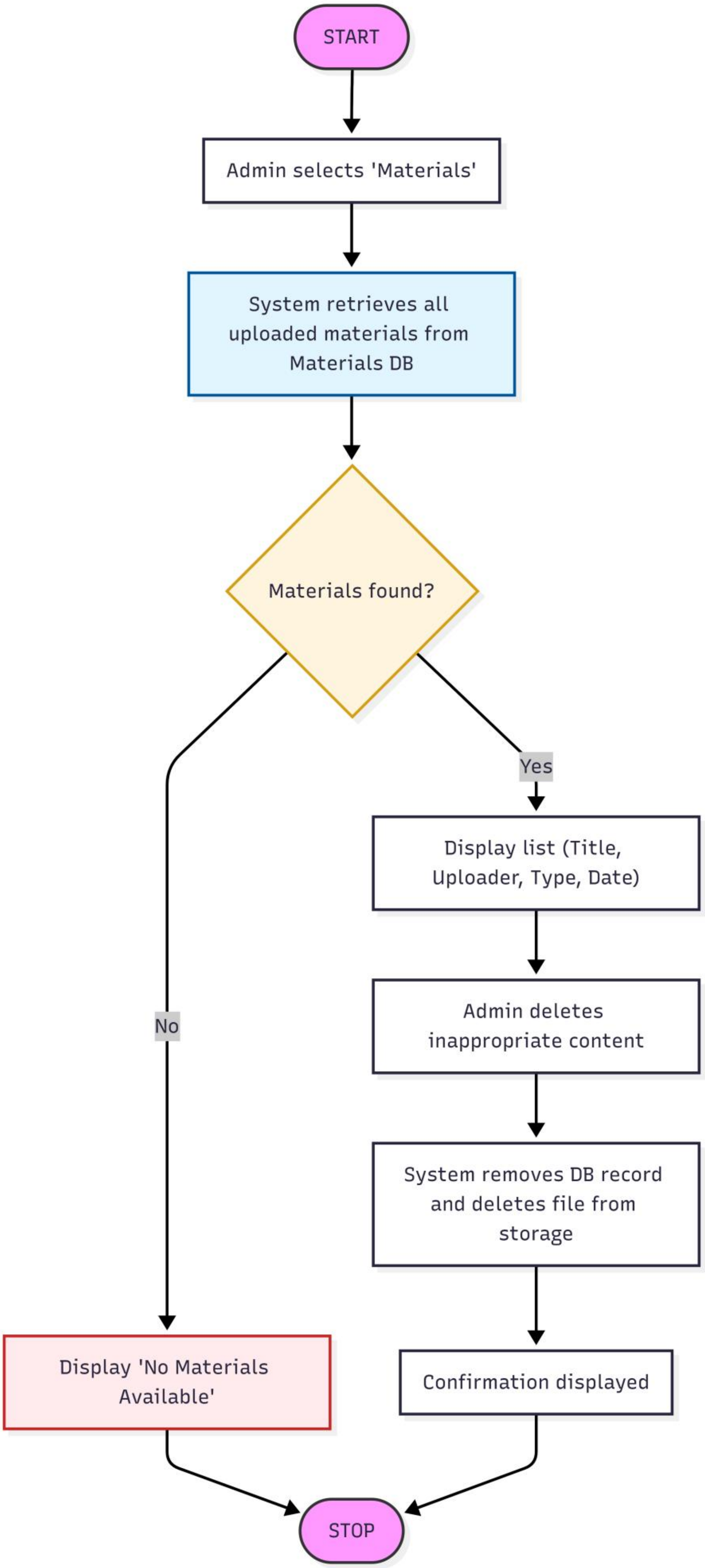


START
Admin selects 'Users' from Dashboard
System queries Users DB for all registered users
Users exist?
No
Yes
Display user list: name, role, status, reg_date
Display 'No Users Available' message
Admin can search or filter users
STOP

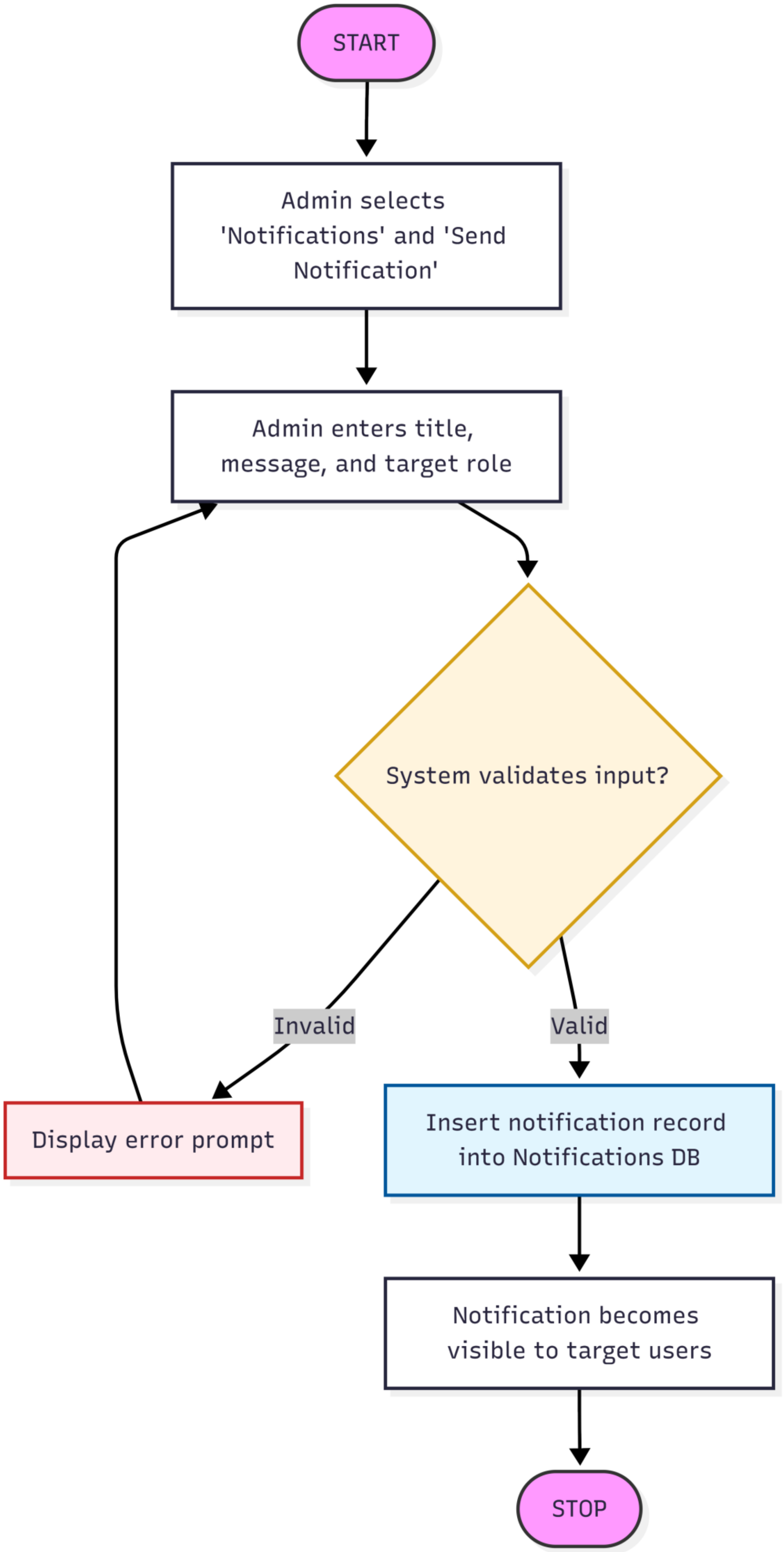
**Manage Approvals: Procedural Details**



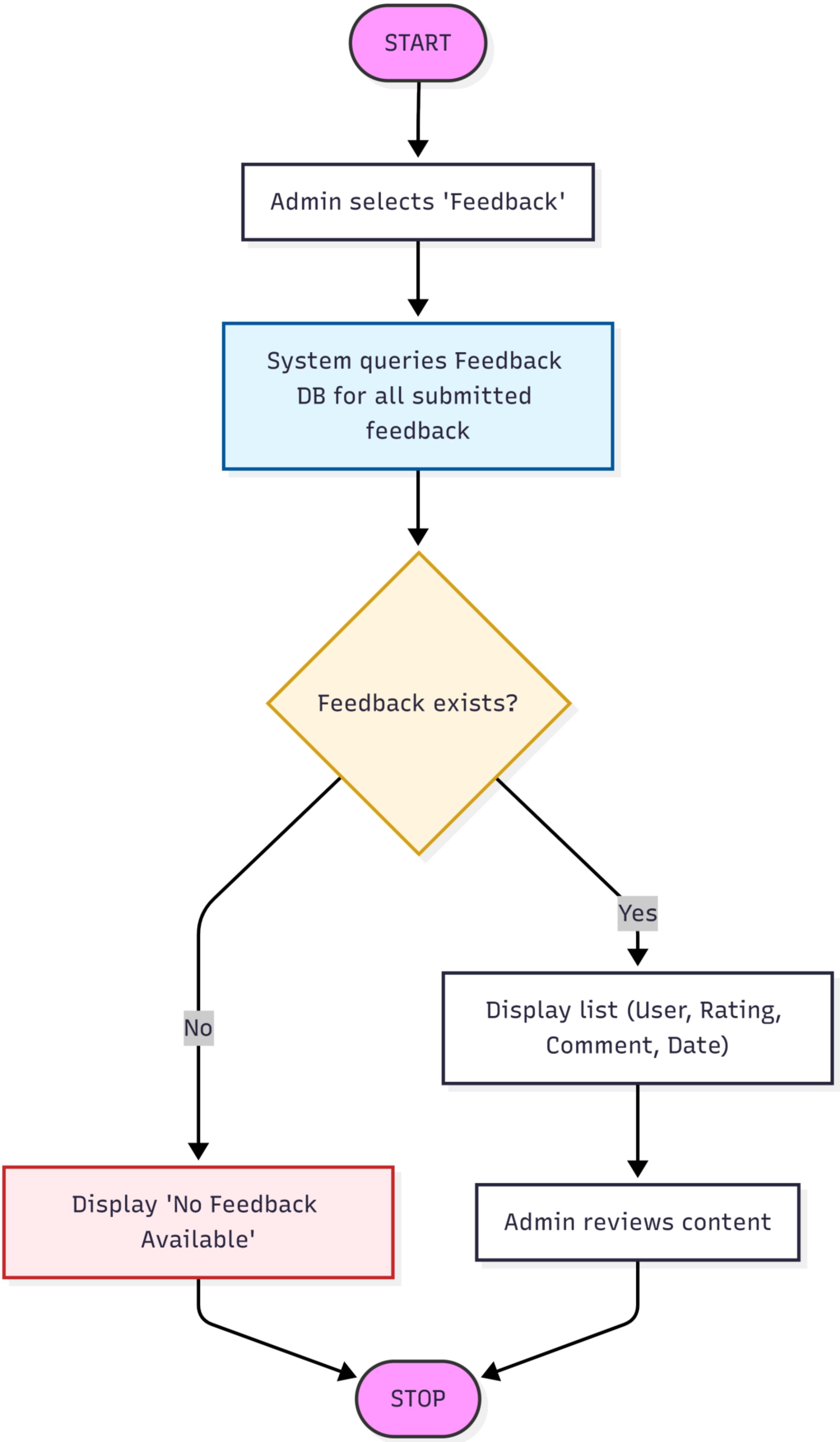
**Manage Materials: Procedural Details**



**Manage Notifications: Procedural Details**



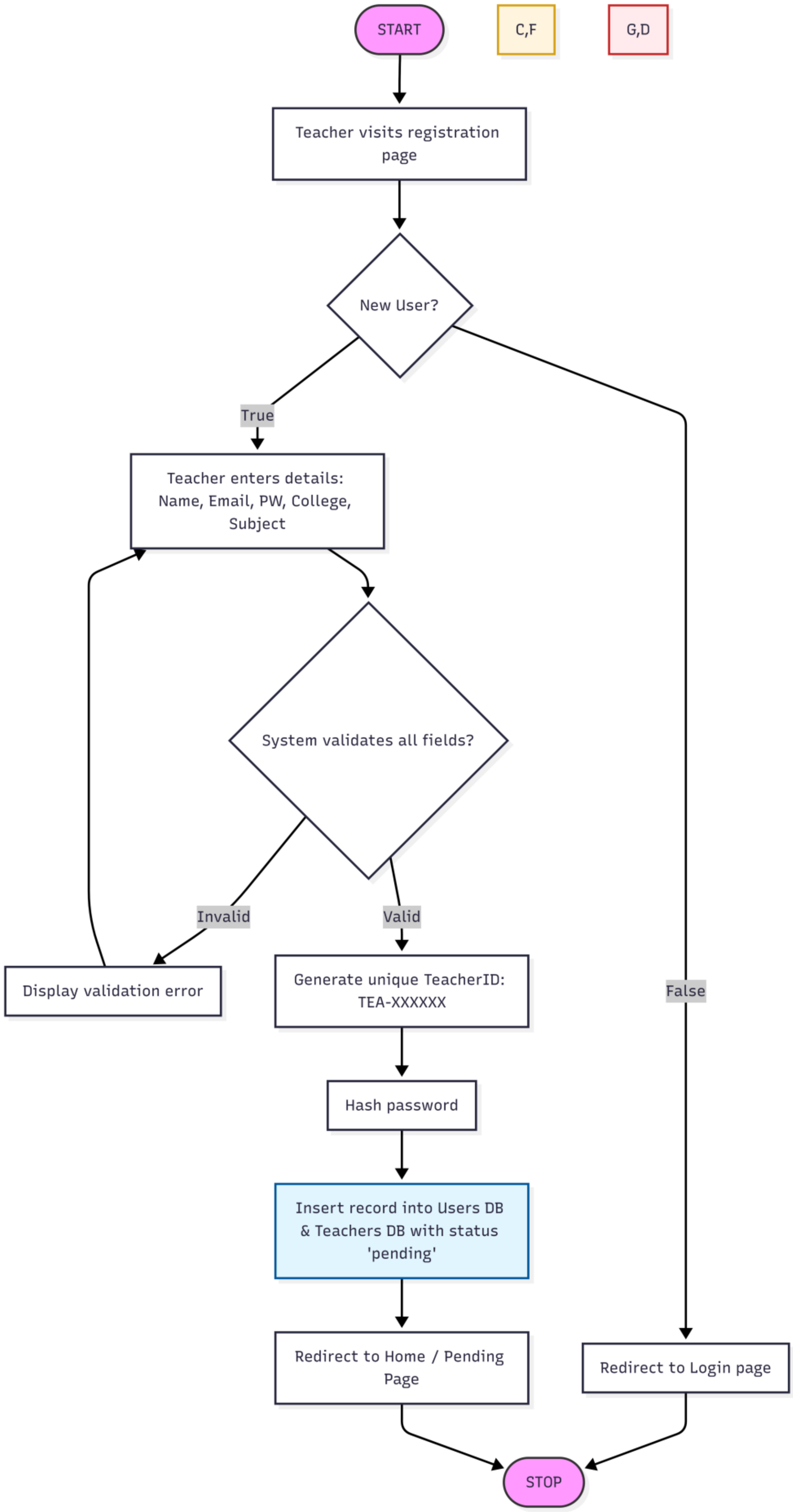
**View Feedback: Procedural Details**



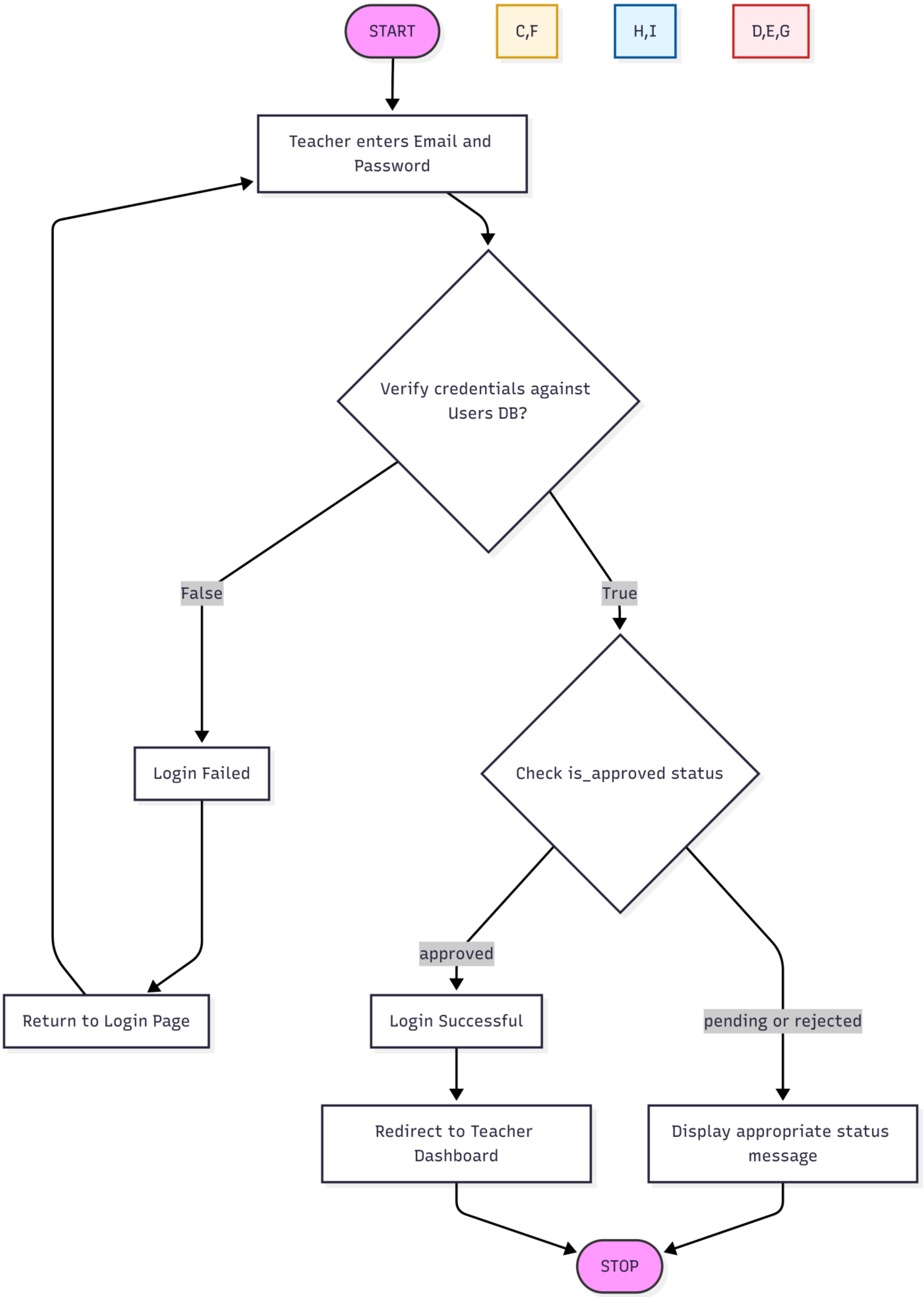

**b.**

## SERVICE PROVIDER (TEACHER):

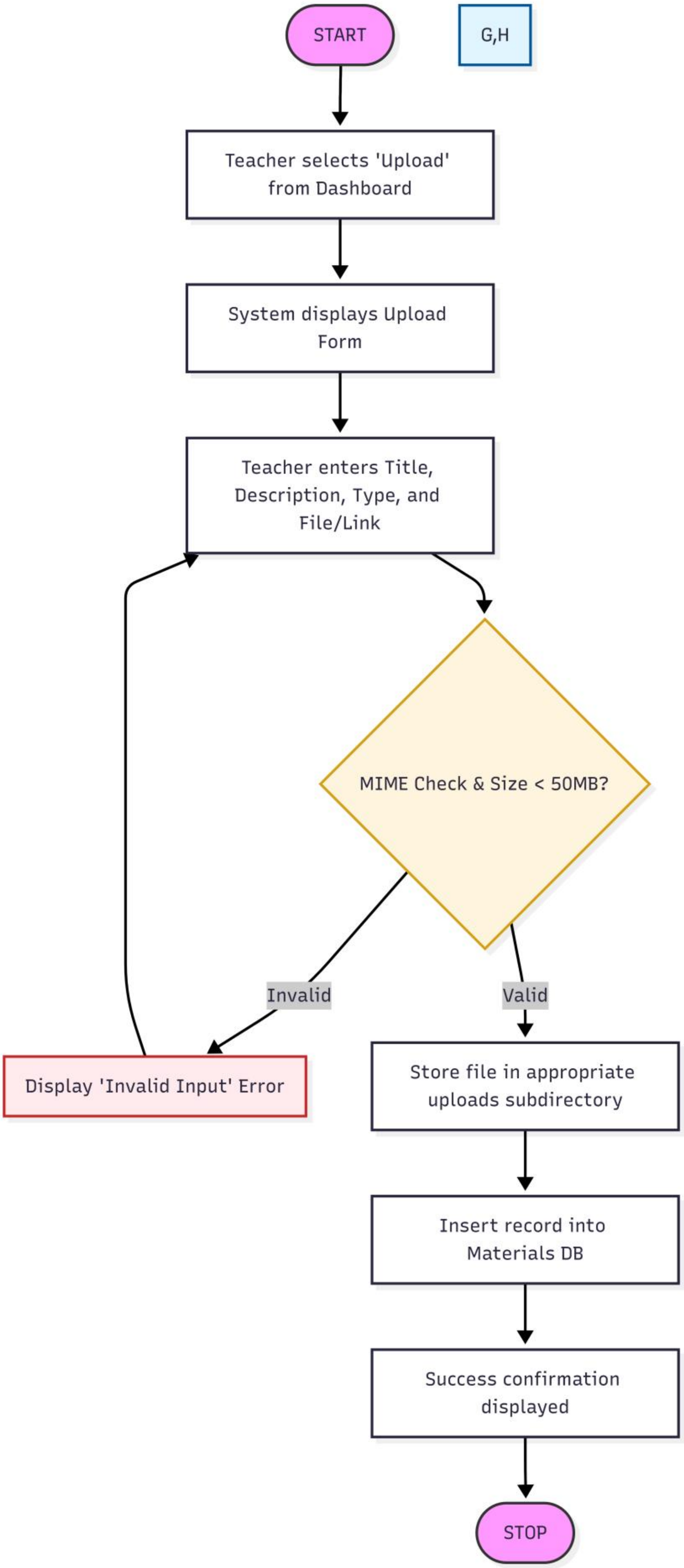
### Registration: Procedural Details



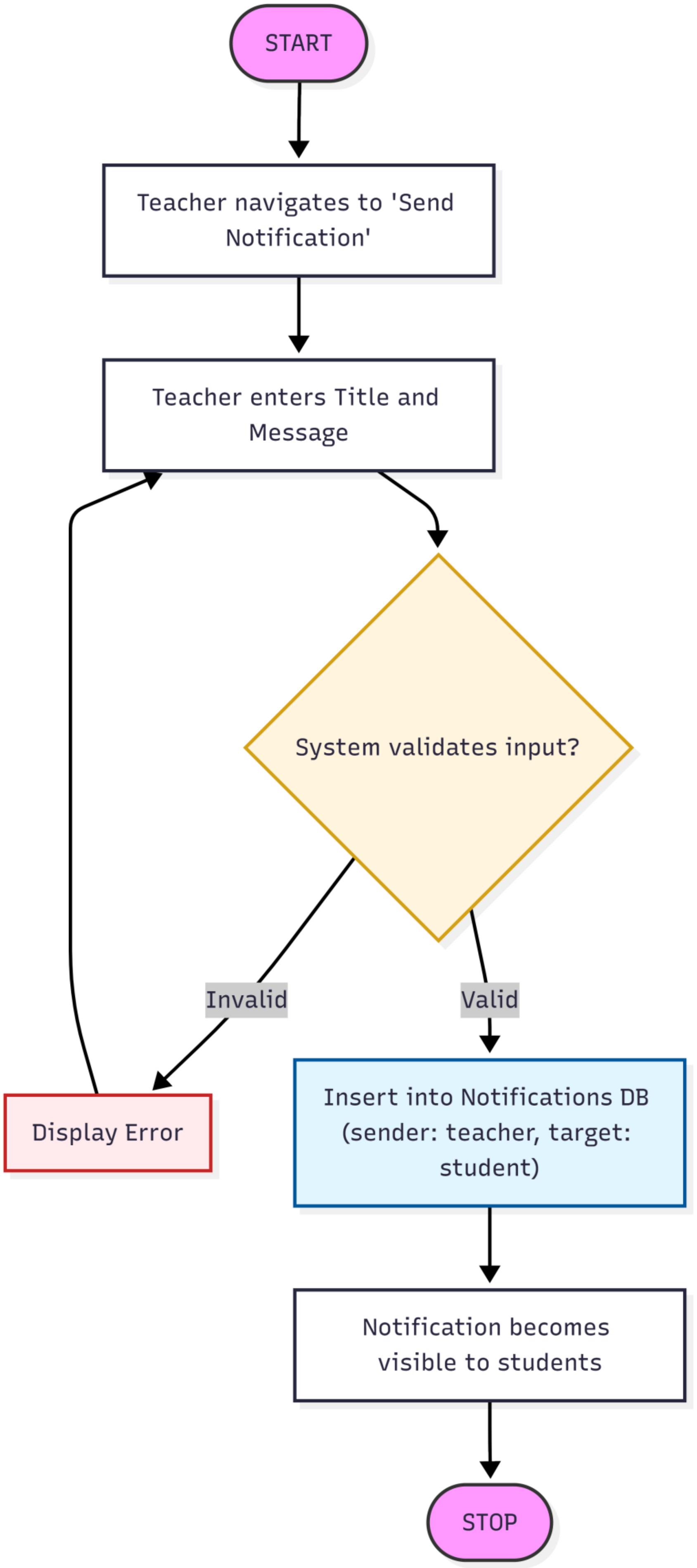
**Login: Procedural Details**



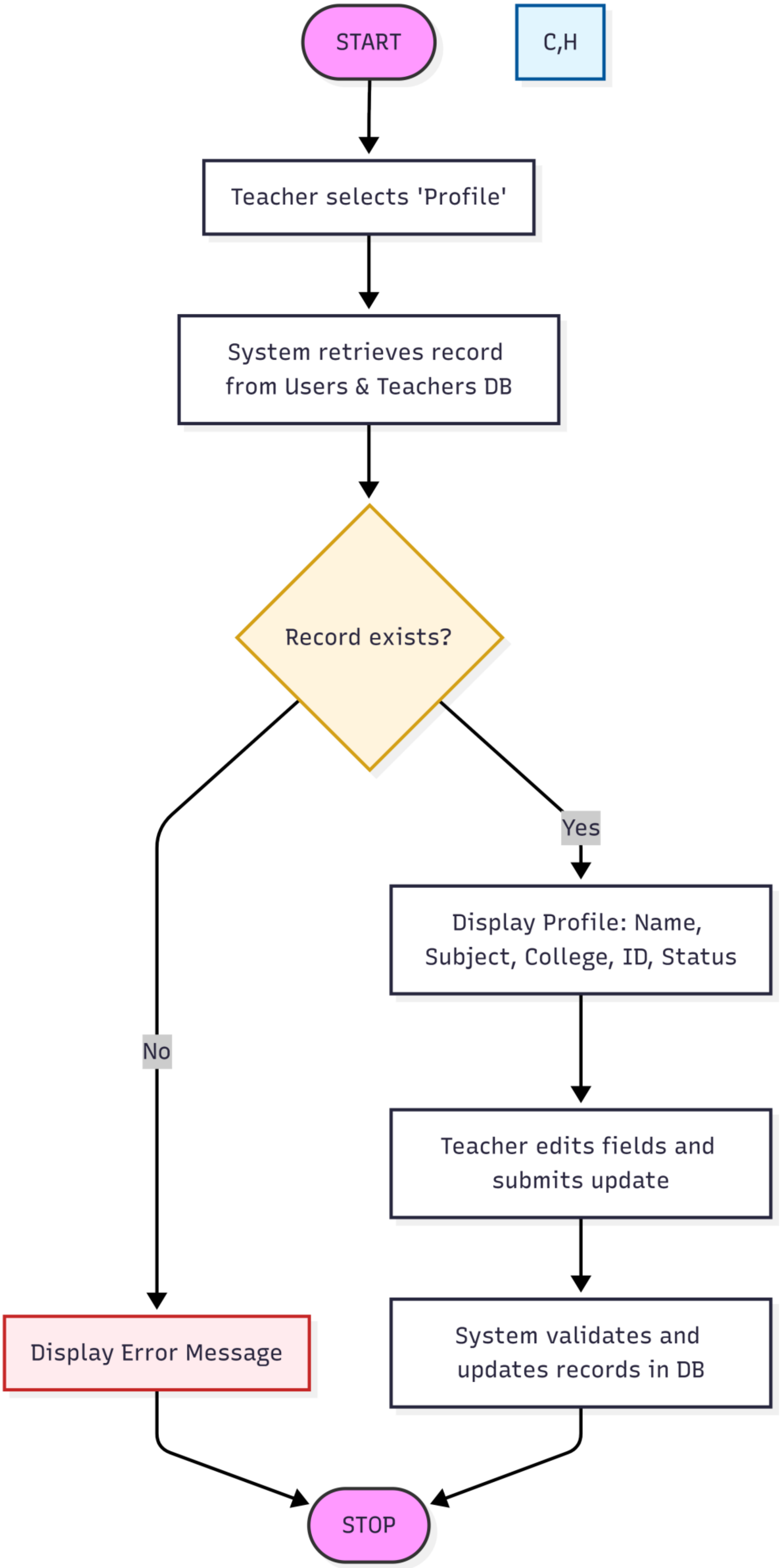
### Upload Materials: Procedural Details



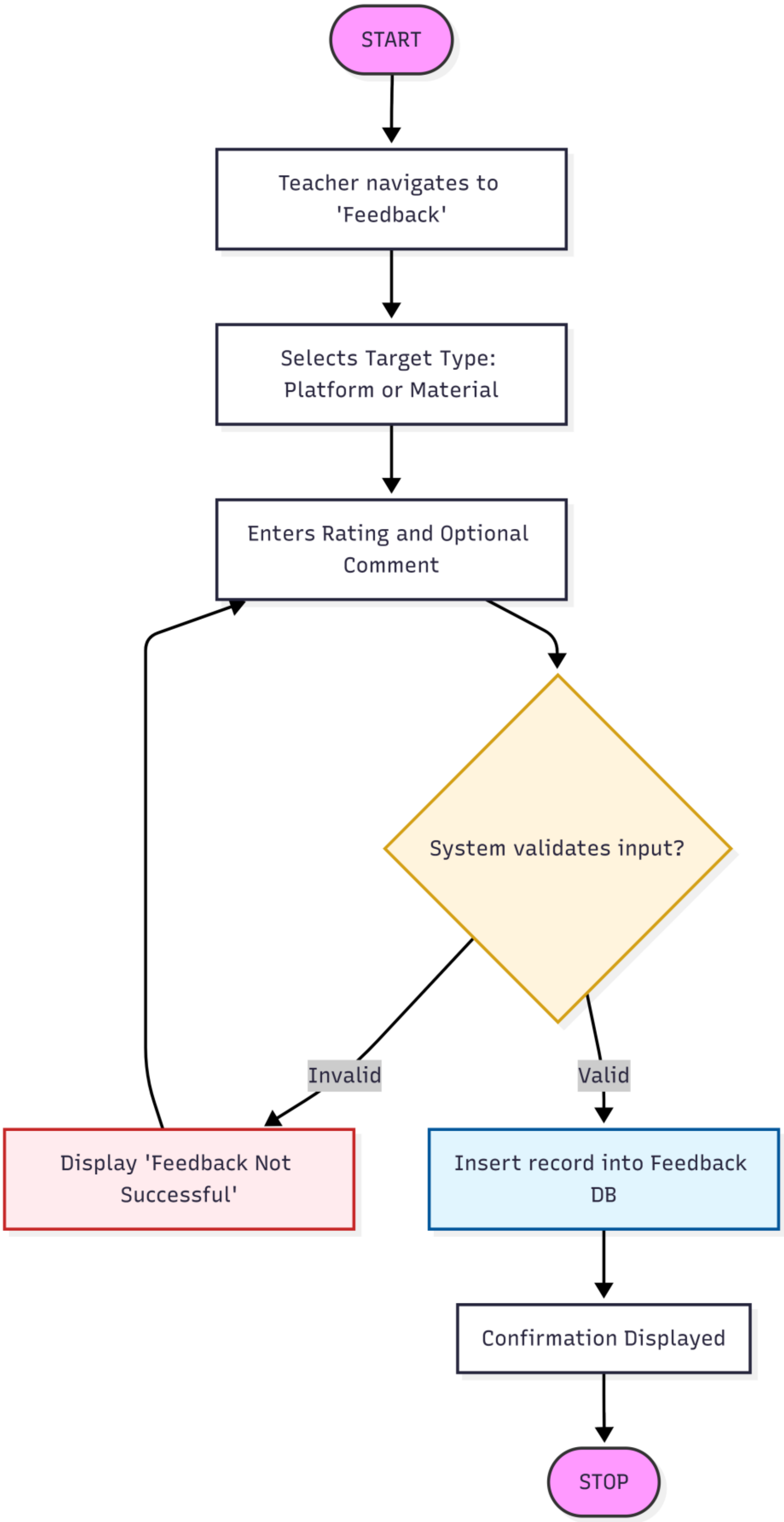
**Send Notification: Procedural Details**



**View Profile: Procedural Details**

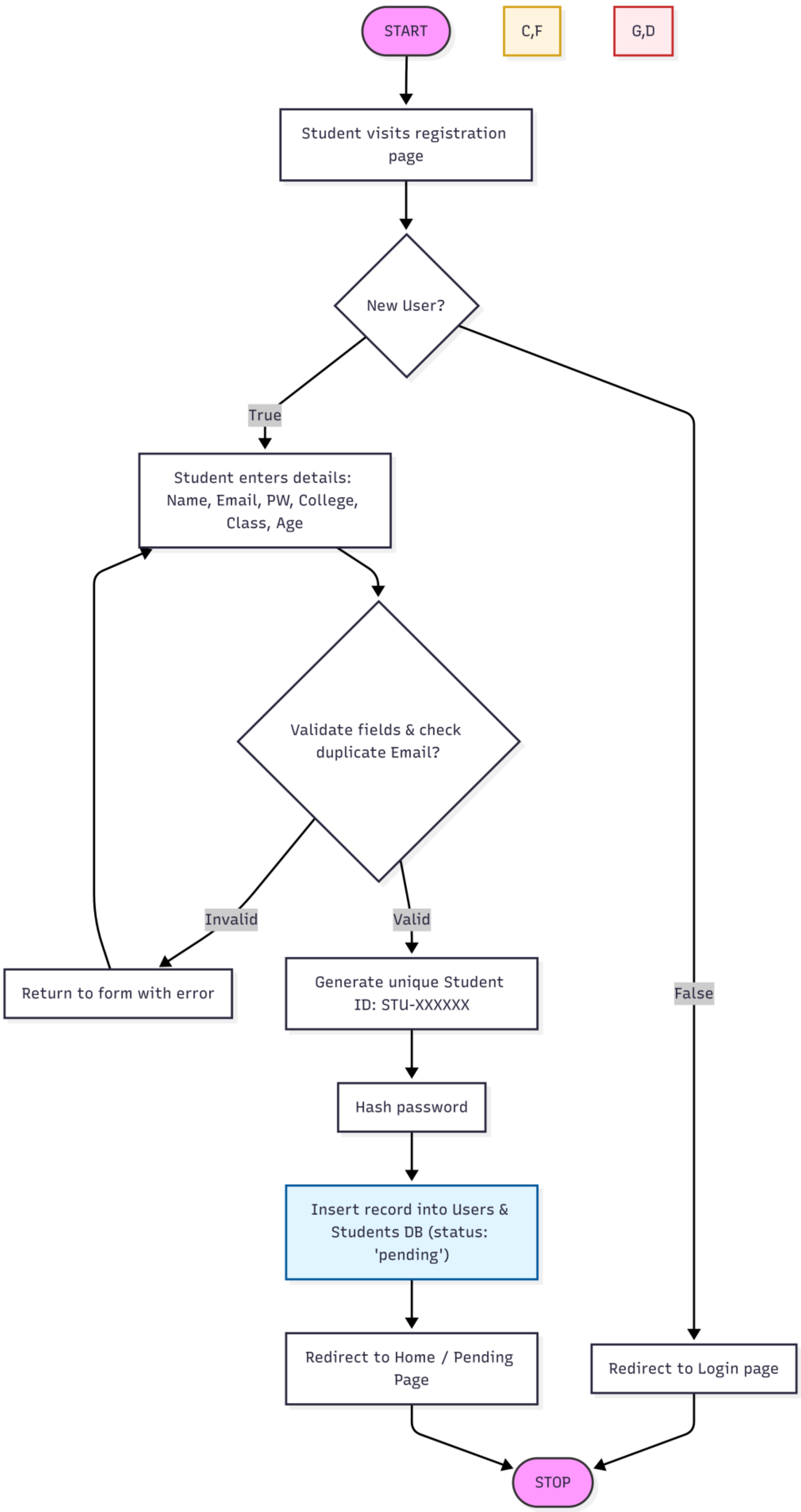


**Submit Feedback: Procedural Details**

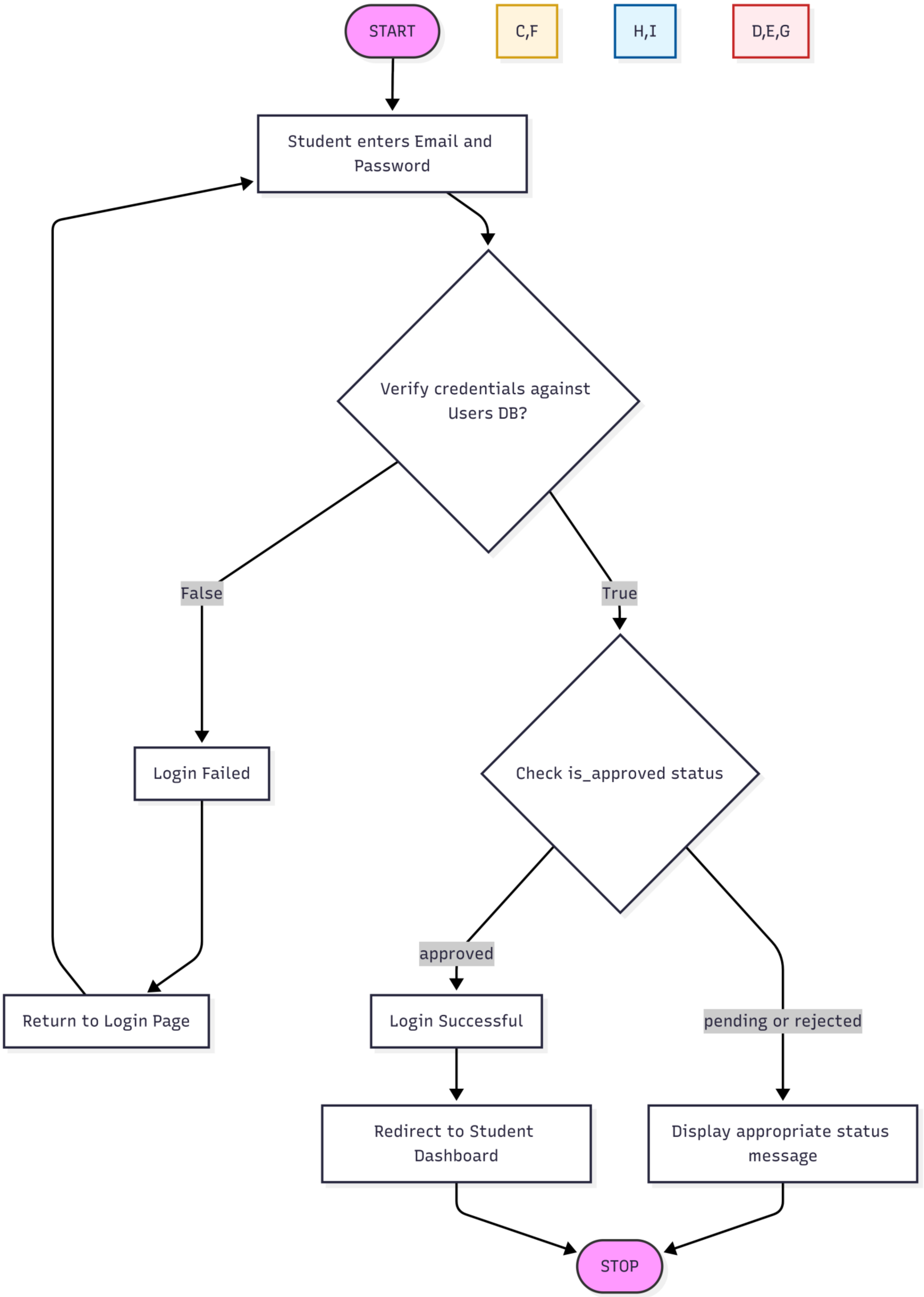


**c. USER (STUDENT):**

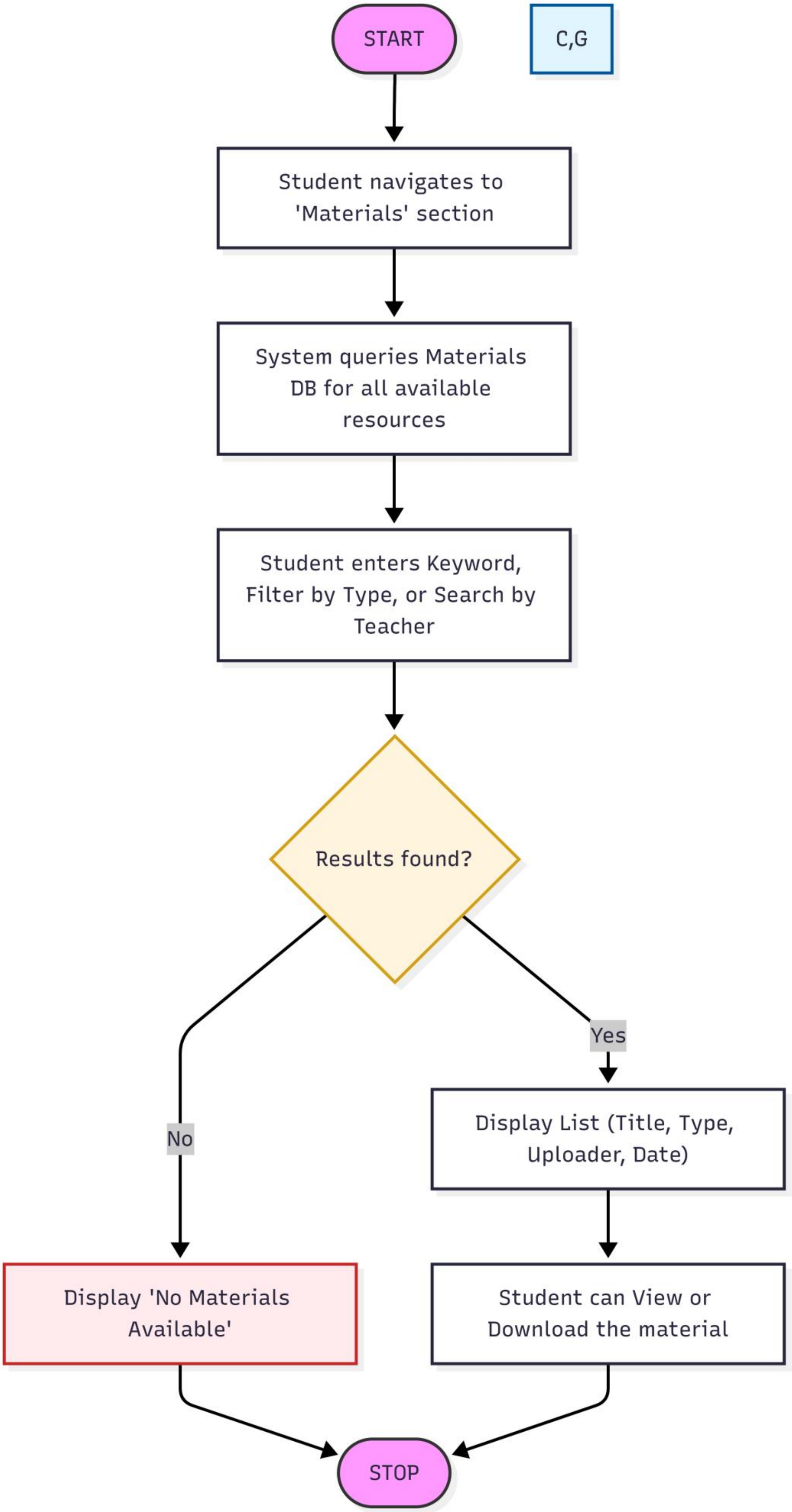
**Register: Procedural Details**



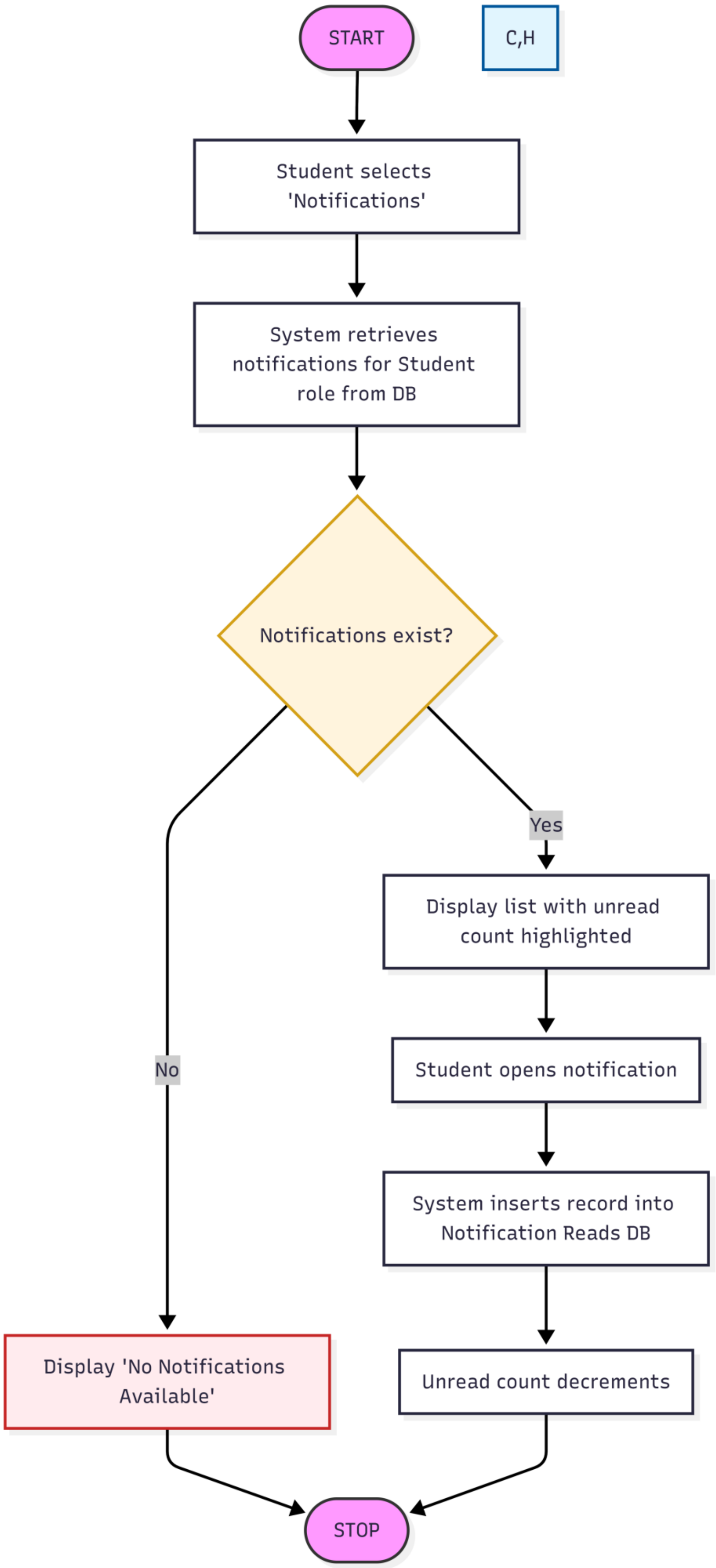
**Login: Procedural Details**



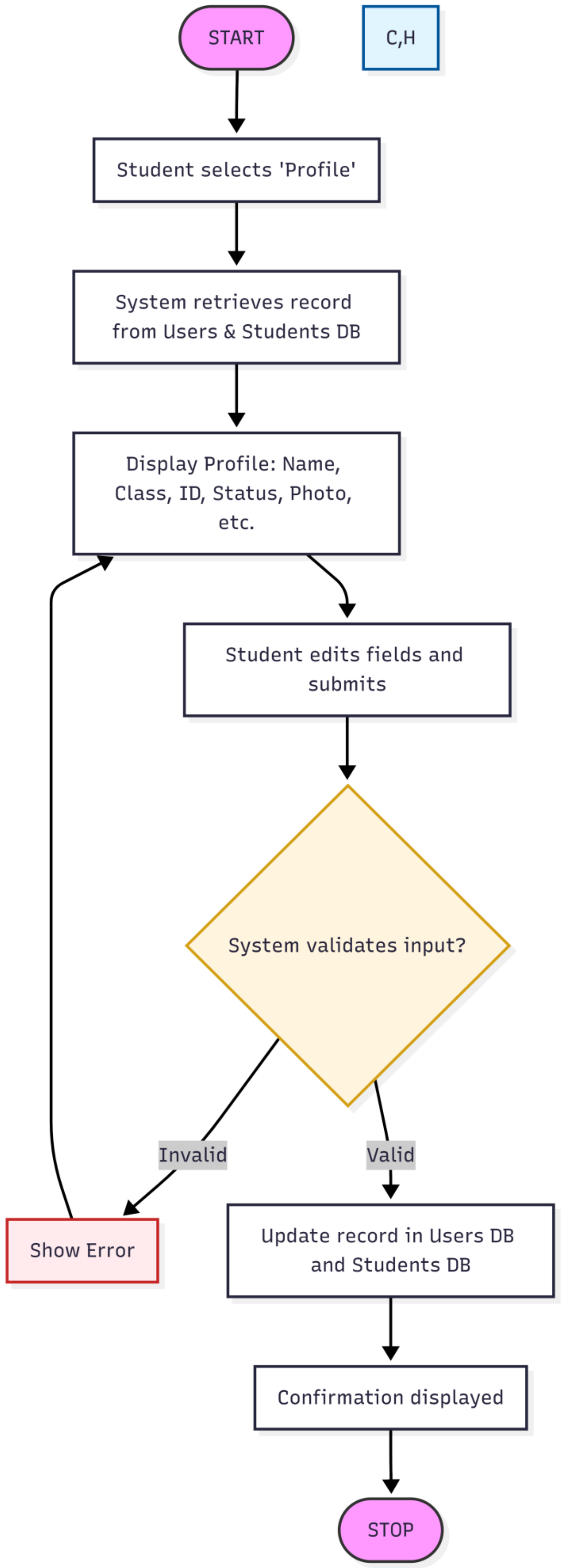
**Browse & Search Materials: Procedural Details**



**View Notifications: Procedural Details**



**View & Update Profile: Procedural Details**



**Submit Feedback: Procedural Details**